TECHNICAL REPORT / ASSIGNMENT SUBMISSION

FastSpeech 2 English and Telugu

Separate acoustic models. Explicit prosody prediction. Reproducible evaluation.

Evaluation package	Contents and provenance
Models / datasets	English: Svarah; Telugu: IndicTTS Telugu. Two independent FastSpeech 2 models.
Selected speech	English step 8,000 / speaker 14; Telugu step 13,000 / speaker 0.
Training record	60,000 logged steps per language; checkpoint selection follows held-out validation minima.
Required outputs	10 test WAVs, 30 duration/F0/energy plots, 4 loss figures.
Implementation	ming024/FastSpeech2, reference d4e79eb; adaptations and paper differences documented below.

Abstract

A phoneme encoder, speaker embeddings, duration/pitch/energy predictors, length expansion and mel decoder form each acoustic model. A pretrained universal HiFi-GAN produces waveforms. This report explains the adaptation, dataset selection, alignment failure recovery, quantitative convergence and every required test input. Saved waveform and predictor measurements are separated from listening judgments. The early validation minima motivate 8k/13k checkpoints rather than assuming the final step is best.

Clickable reading guide

02 / Architecture and equations

03 / Language adaptations and reference differences

04 / Data, alignment and training workflow

05 / English losses and numerical convergence

07 / Telugu losses and numerical convergence

09 / Ten test-sentence analyses: pages 9-18

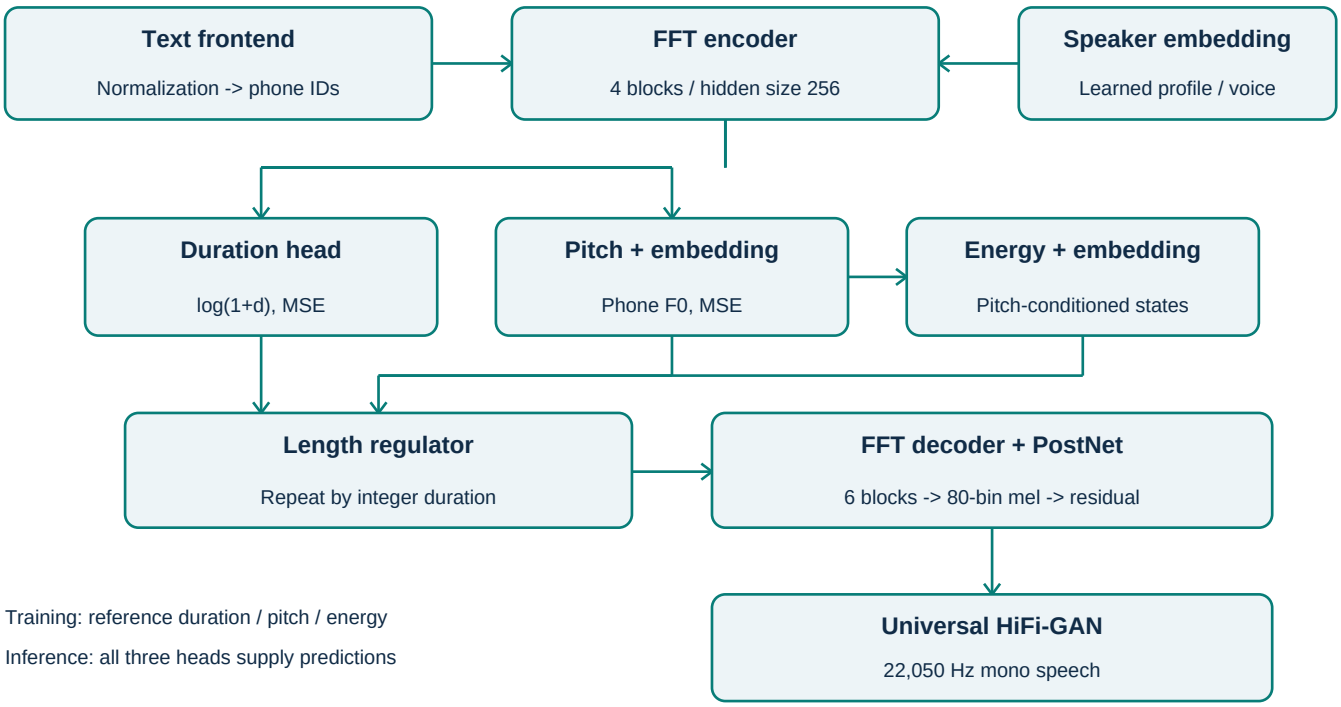
19 / Comparison, failures and improvements

20 / Reproduction commands and model links

Evidence status

Structural checks pass; 4 evidence items remain open. Local selected checkpoint bytes match the registry. Reserve audio-hash coverage and per-sentence listening observations are incomplete. Source snapshots and measured results are provided; no MOS, WER, naturalness score or successful retraining is invented.

1 / Architecture and mathematical formulation



Training: reference duration / pitch / energy

Inference: all three heads supply predictions

Training and synthesis

For token i, the duration head predicts z_i with target log(1 + d_i). At default inference controls, the expansion count is d_hat_i = max(round(exp(z_i) - 1), 0), and output length is T = sum_i d_hat_i. Training supplies aligned reference durations and reference pitch/energy; free-running inference supplies predicted values. A low reference-conditioned mel loss does not establish free-running intelligibility.

Loss and feature definitions

L = L1(mel, target) + L1(postnet_mel, target) + MSE(z, log(1+d)) + MSE(pitch, pitch_target) + MSE(energy, energy_target). Padding masks exclude padded tokens/frames. Energy is the L2 norm of linear STFT magnitudes, not dB SPL. F0 is extracted with PyWorld and, in this baseline, interpolated and averaged per phone before normalization.

Setting	Committed baseline
FFT dimensions	4 encoder / 6 decoder blocks; hidden 256; 2 attention heads; convolution filter 1024
Predictors	Two convolution layers; kernel 3; 256 channels; dropout 0.5
Variance embeddings	256 linear bins for normalized phone-level pitch and energy
Output	80 mel channels plus residual PostNet; separate pretrained HiFi-GAN

2 / Reference reuse and language adaptations

Component	Adaptation / implication
Reference Transformer	Encoder/decoder FFT structure retained from ming024. The project is an adaptation, not a bit-for-bit reproduction.
Variance adaptor	Energy-control routing corrected; length-regulator tensors use the input device. Do not label model/modules.py unmodified.
English frontend	Lexicon lookup with g2p_en OOV fallback; time/date/number expansion. First-pronunciation lookup can miss noun/verb stress distinctions.
Telugu frontend	Deterministic Unicode decomposition: consonant plus inherent vowel unless a vowel sign or virama changes it; dependent signs mapped to independent vowel phones.
Speaker conditioning	Both configurations enable learned speaker embeddings. English demographic groups are not independently verified individual identities.
Preprocessing	Interior silence preservation, timeline checks, token checks and selected-training-only normalization/bounds.
Training / inference	CLI paths and explicit checkpoint overrides; caching, checkpoint resume and prediction summaries.

Differences from the supplied v8 paper

Aspect	Paper v8	Project baseline
Decoder depth	4 FFT blocks	6 FFT blocks
Pitch target	Frame-level CWT	Direct phone-level F0
Energy target	Frame-level	Phone-level
Added PostNet	Absent	Residual convolutional PostNet
Vocoder	Parallel WaveGAN	Universal HiFi-GAN

Telugu representation limits

The Unicode frontend is a phonemic approximation. Anusvara assimilation, loanwords and allophones remain limitations; determinism alone does not establish linguistic validity. Symbol inventories are preserved with the model metadata.

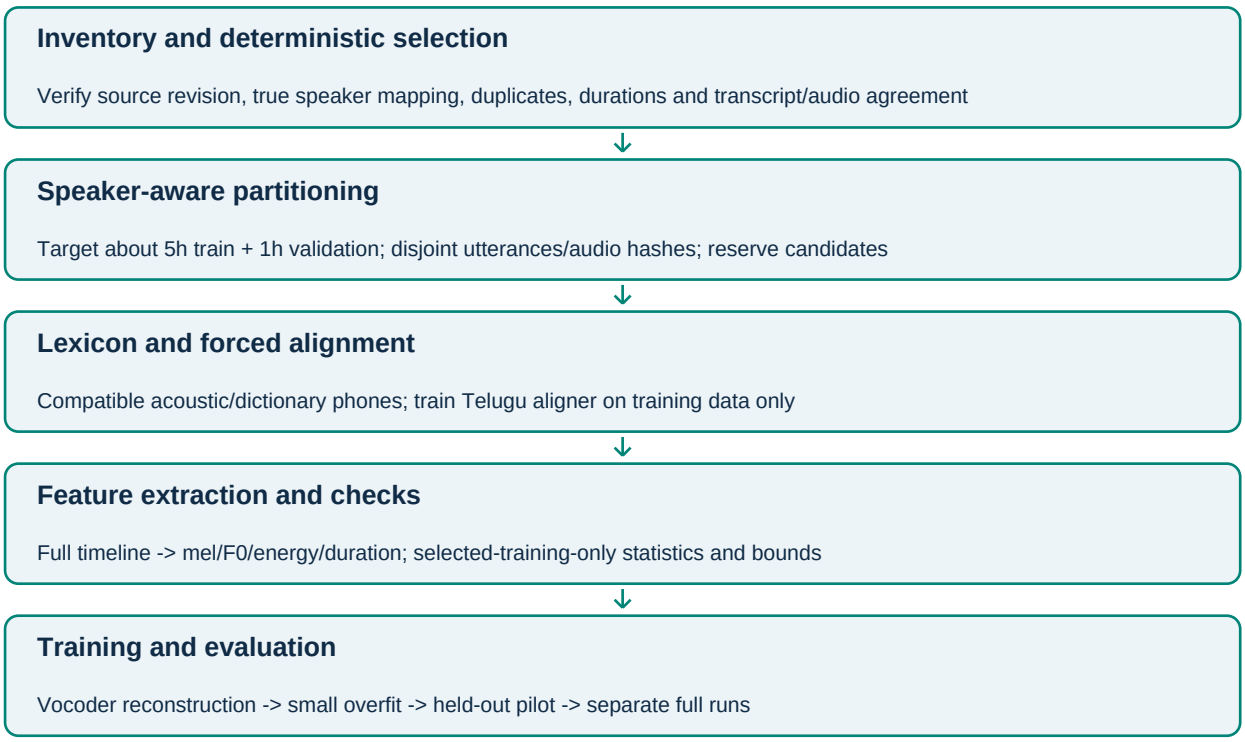
Alignment and OOV handling

Review found 82 historical Telugu target/frontend mismatches (73 training, 9 validation), often around dotted initials. Preparation now shares inference word boundaries, and new preprocessing rejects unknown-speech spn instead of labeling it silence. Old targets and audio remain historical evidence; these code fixes do not retrain existing weights.

Normalization and test fidelity

The frozen UTF-8 test manifest preserves Telugu diacritics and E3/T3 double spaces. Normalization is recorded separately; source spacing does not establish an audible pause.

3 / Data, alignment and experiment protocol



Selection implemented in the committed preparation scripts

Both scripts default to seed 42, 1-20 second clips, mono 22,050 Hz PCM output, waveform SHA-1 deduplication and speaker-stratified utterance splits. English creates demographic profiles from eight fields; Telugu uses both female and male labels. Split-specific reserves support replenishment after alignment or feature rejection. These defaults describe code, not independently verified final-run settings.

Required dataset evidence

The target is approximately five training hours and one validation hour per language after trimming and rejection. Recovered preprocessing summaries report: English retained 3,359 training utterances (5.000h) across 115 speakers and 674 validation utterances (1.001h) across 114 speakers; Telugu retained 2,164 training utterances (5.000h) and 421 validation utterances (1.000h) across 2 studio speakers. The published ID-only manifests confirm zero ID overlap and trained-speaker coverage. Final waveform SHA-1 evidence is incomplete for reserve clips; source preparation statistics separately report zero hash overlap.

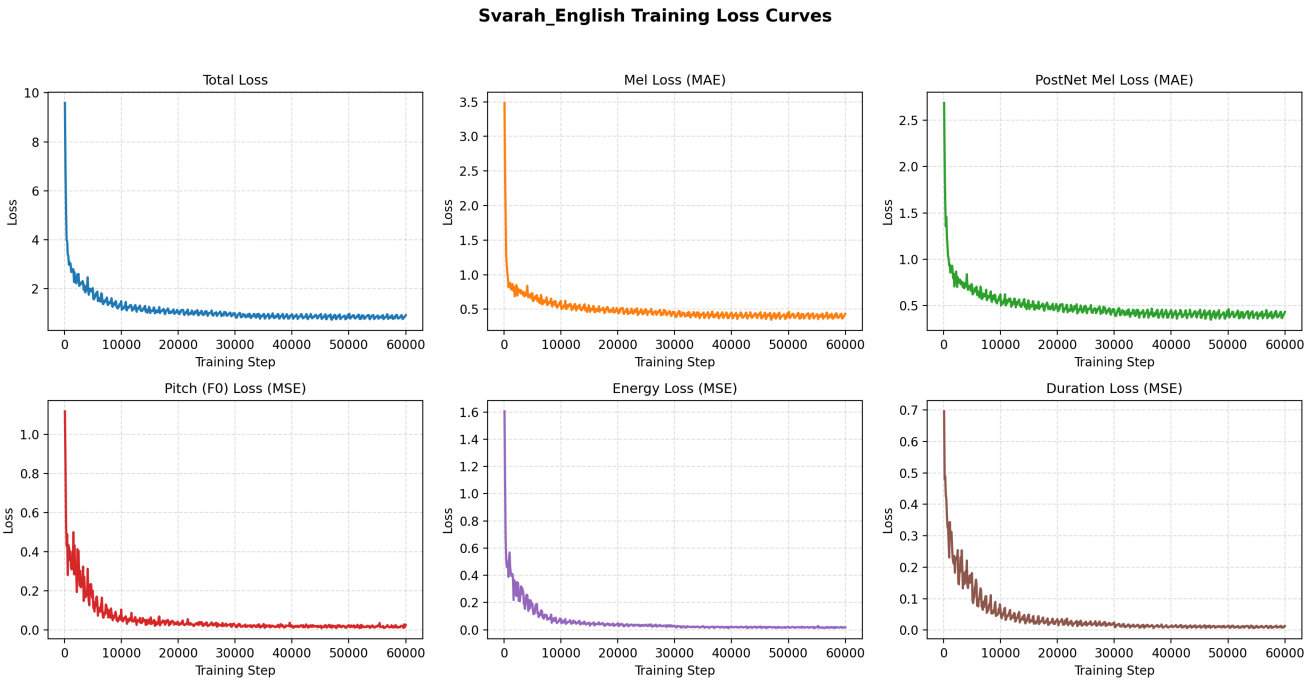
Alignment failure and its correction

The earlier reader omitted empty interior pause intervals while audio trimming retained those spans. Subsequent phone boundaries then mapped to earlier frames. The fix reads empty intervals, maps interior silence/gaps to sp, rejects overlaps and checks the pre-slice time span. Equality after slicing is insufficient: a truncated target can still equal an incorrect duration sum.

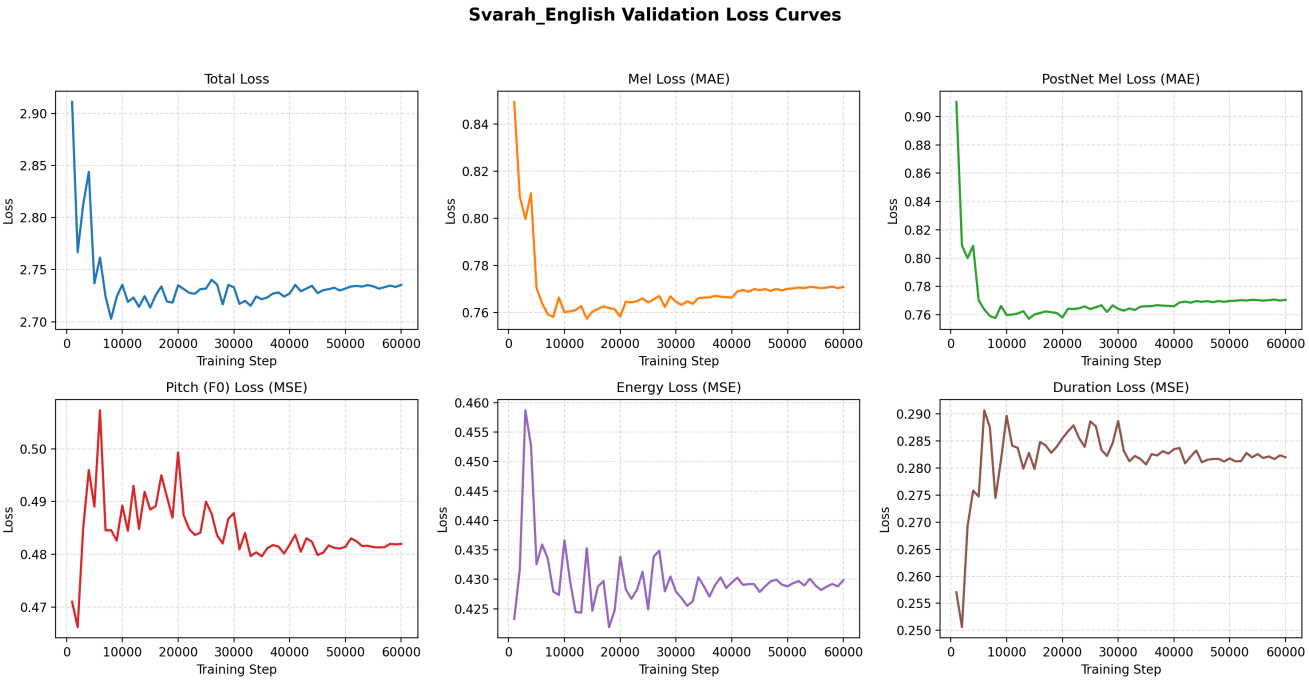
Optimization and checkpoint distinction

Committed defaults: batch 16; Adam betas (0.9, 0.98), epsilon 1e-9; warmup 4,000; clipping 1.0; total 60,000; annealing at 30k/45k/55k. The output metadata identifies 8k/13k checkpoints; the supplied CSV histories independently place the lowest validation totals at those steps. The historical environment and elapsed time remain unavailable. New runs save configuration, dataset revision and feature/source fingerprints; resume refuses changed targets.

4 / English: loss-curve analysis



Training: total and five component losses



Validation: total and five component losses

Visual interpretation

The validation total improves early, reaches its lowest visible region near the early checkpoints, then fluctuates around a plateau. Mel and PostNet validation errors gradually rise after their early decrease while training errors continue falling. This widening generalization gap is consistent with overfitting; more optimizer steps alone are not a remedy.

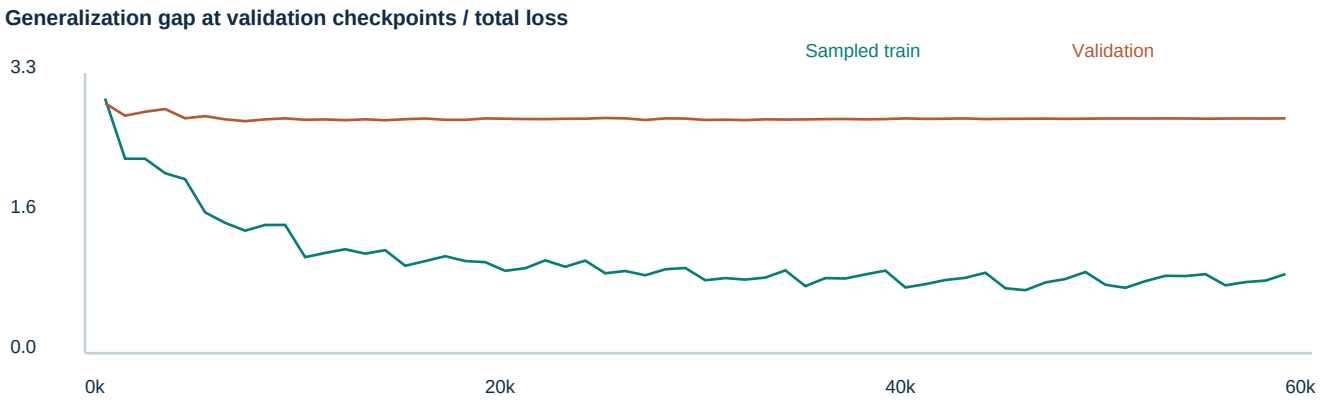
The figures extend to 60,000 steps. The following table is computed from the preserved CSVs. Training values are sampled minibatch losses; validation values are the trainer's aggregate, so their difference is descriptive rather than a matched-sample statistical estimate.

4 / English: numerical convergence evidence

Split	Step	Total	Mel	PostNet	Pitch	Energy	Duration
Train: first	100	9.5892	3.4841	2.6863	1.1178	1.6051	0.6959
Train: selected	8000	1.4274	0.5931	0.5930	0.0854	0.0776	0.0783
Train: final	60000	0.9216	0.4331	0.4326	0.0259	0.0168	0.0131
Val: first	1000	2.9111	0.8495	0.9103	0.4710	0.4233	0.2570
Val: minimum	8000	2.7028	0.7581	0.7577	0.4846	0.4279	0.2745
Val: final	60000	2.7353	0.7709	0.7706	0.4820	0.4299	0.2820

There are 600 training and 60 validation records. The lowest logged validation total is 2.702820 at step 8,000, a 7.15% decrease from the first logged validation value. The saved audio metadata selects step 8,000, which matches this minimum.

Selection and overfitting

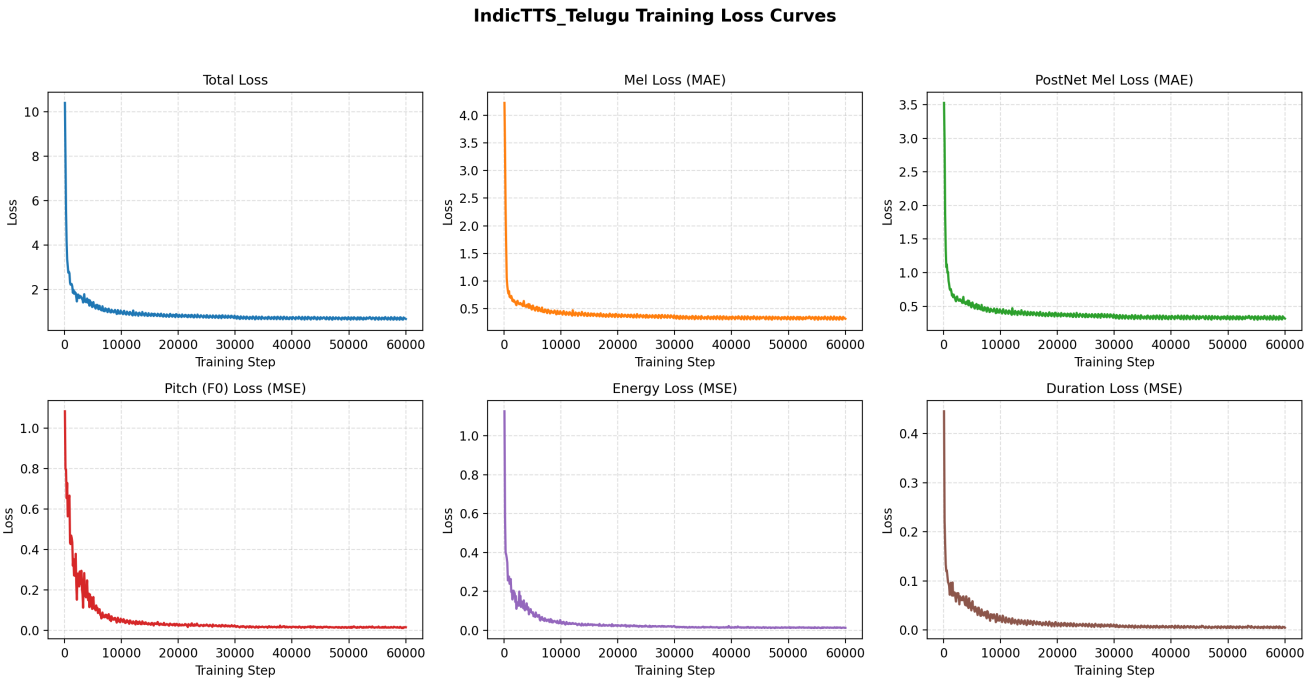


The final checkpoint is not automatically the best generalizing checkpoint. The continuing reduction in training error alongside relatively flat or rising validation errors supports early checkpoint selection. The chosen step has the best logged loss, but perceptual quality still requires free-running listening evaluation. Neither total loss nor a statistically valid duration array establishes correct articulation.

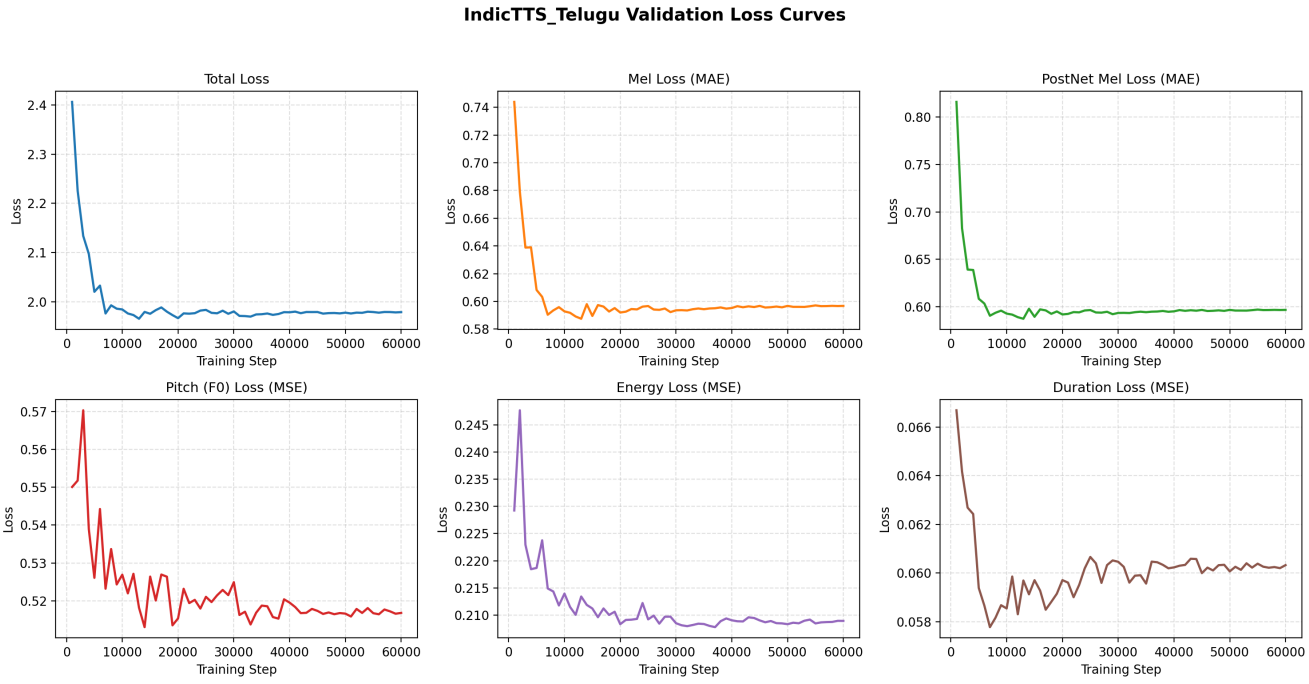
Log provenance

Unmodified CSVs are preserved under Results/English/metrics/. The audit records their presence, and the report derives tables directly from them. The CSVs do not record all dataset, environment, checkpoint-hash or listening information; these must be supplied separately. Losses from different corpora should not be treated as directly comparable naturalness scores.

4 / Telugu: loss-curve analysis



Training: total and five component losses



Validation: total and five component losses

Visual interpretation

Training losses fall substantially over the run. Validation improves early and then largely plateaus, with a persistent train/validation gap. Compare each component on its own scale: raw totals sum heterogeneous losses. Lower Telugu losses do not by themselves establish better intelligibility or a causal language advantage.

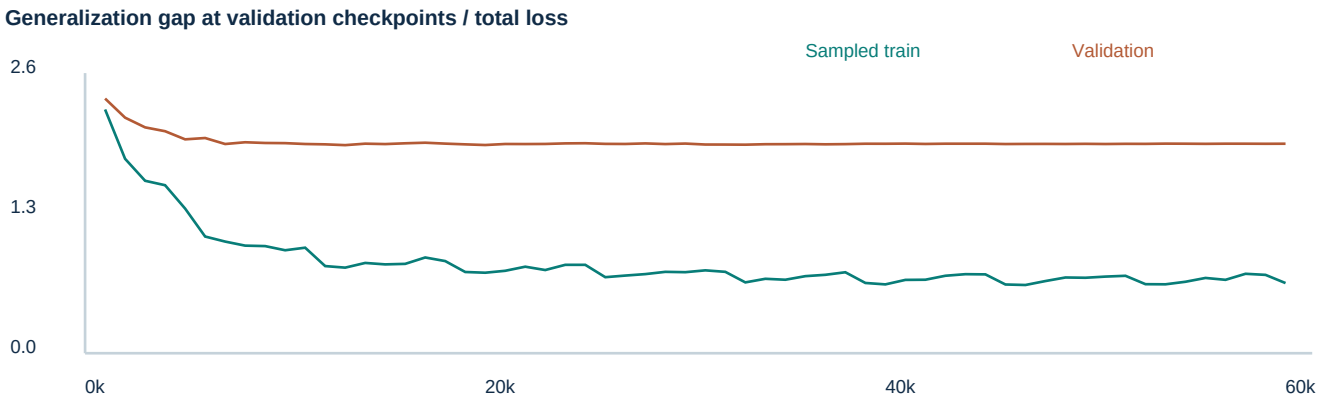
The figures extend to 60,000 steps. The following table is computed from the preserved CSVs. Training values are sampled minibatch losses; validation values are the trainer's aggregate, so their difference is descriptive rather than a matched-sample statistical estimate.

4 / Telugu: numerical convergence evidence

Split	Step	Total	Mel	PostNet	Pitch	Energy	Duration
Train: first	100	10.4081	4.2251	3.5288	1.0831	1.1258	0.4454
Train: selected	13000	0.8086	0.3687	0.3684	0.0321	0.0284	0.0111
Train: final	60000	0.6627	0.3153	0.3150	0.0152	0.0126	0.0045
Val: first	1000	2.4060	0.7439	0.8161	0.5500	0.2292	0.0667
Val: minimum	13000	1.9660	0.5875	0.5872	0.5182	0.2134	0.0597
Val: final	60000	1.9794	0.5968	0.5966	0.5168	0.2089	0.0603

There are 600 training and 60 validation records. The lowest logged validation total is 1.965954 at step 13,000, a 18.29% decrease from the first logged validation value. The saved audio metadata selects step 13,000, which matches this minimum.

Selection and overfitting



The final checkpoint is not automatically the best generalizing checkpoint. The continuing reduction in training error alongside relatively flat or rising validation errors supports early checkpoint selection. The chosen step has the best logged loss, but perceptual quality still requires free-running listening evaluation. Neither total loss nor a statistically valid duration array establishes correct articulation.

Log provenance

Unmodified CSVs are preserved under Results/Telugu/metrics/. The audit records their presence, and the report derives tables directly from them. The CSVs do not record all dataset, environment, checkpoint-hash or listening information; these must be supplied separately. Losses from different corpora should not be treated as directly comparable naturalness scores.

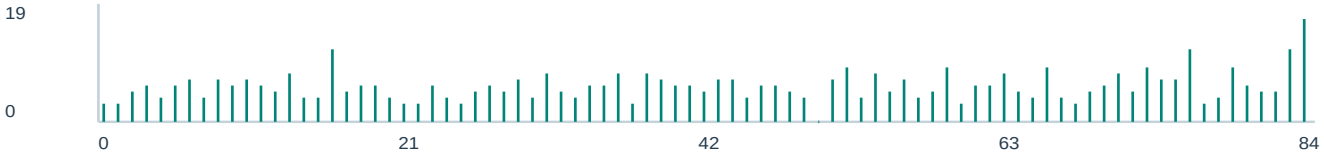
5 / E1: per-sentence prosody analysis

English | step 8,000 | speaker ID 14

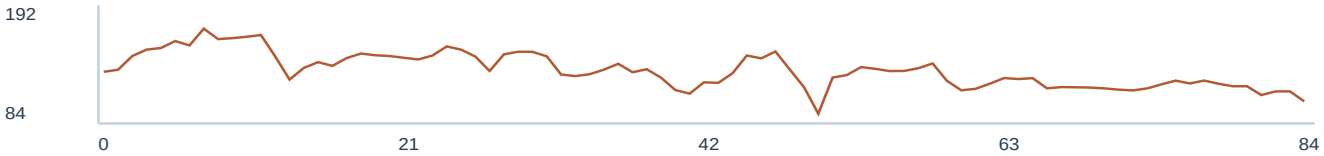
The antidisestablishmentarianism movement gained considerable momentum during the parliamentary proceedings.

Saved measurement	Value
Tokens / expansion / duration	85 / 501 frames / 5.817 seconds
F0 token mean [range]	134.65 [93.11, 174.44] Hz
Energy token mean [range]	77.90 [13.95, 149.04]
Zero-duration tokens / explicit pause	1 tokens / 0 frames (0.000 s)

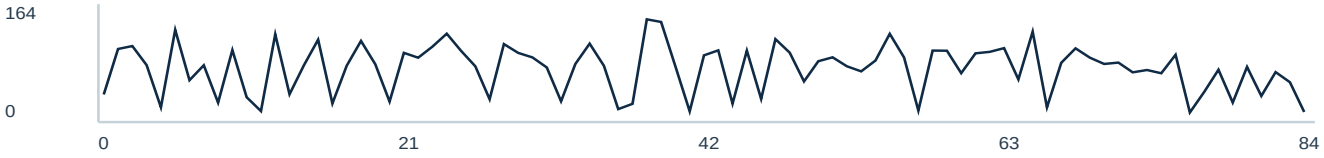
Duration / mel frames



Predicted continuous F0 / Hz



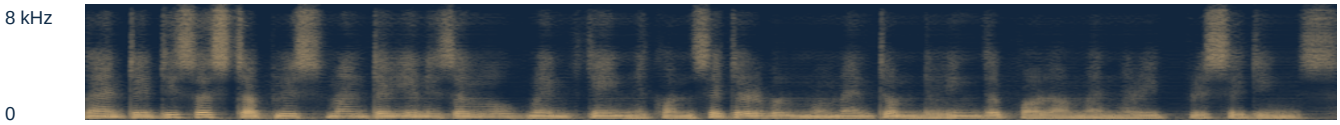
Predicted energy / STFT-magnitude units



x-axis: token index; original labeled plots accompany the submission



Measured waveform and spectrogram (Hann 1024 / hop 128; dB relative to utterance peak)



Waveform measurement: RMS -16.8 dBFS; peak -3.2 dBFS; 0 full-scale samples. The spectrogram describes generated acoustic structure and does not establish correct words.

The rare long word stresses pronunciation generation and sentence completeness. The saved sequence contains a zero-duration token; a zero expansion removes that token from the frame sequence. This merits listening review but does not alone prove a missing audible word.

Listening observation: pending. A reviewer must assess pronunciation, completeness and prosody for this exact WAV. Predicted continuous F0 includes interpolated unvoiced regions; token means are unweighted.

Open E1 audio | Raw text and prediction arrays

5 / E2a: per-sentence prosody analysis

English | step 8,000 | speaker ID 14

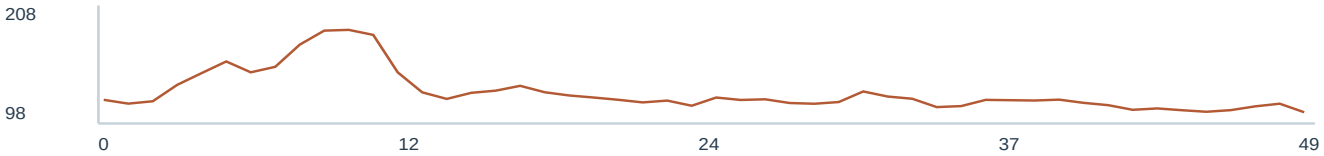
The orchestra's conduct during the ceremony was praised by the audience.

Saved measurement	Value
Tokens / expansion / duration	50 / 283 frames / 3.286 seconds
F0 token mean [range]	128.44 [109.06, 189.23] Hz
Energy token mean [range]	68.40 [7.16, 138.10]
Zero-duration tokens / explicit pause	0 tokens / 0 frames (0.000 s)

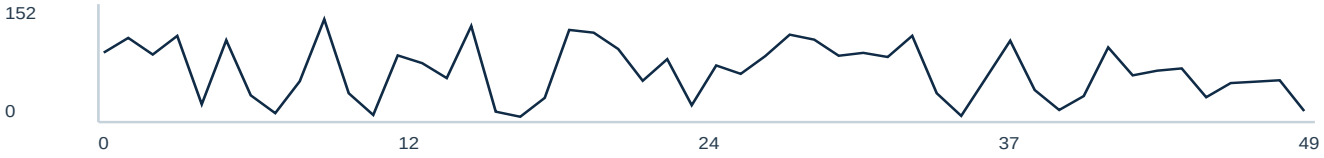
Duration / mel frames



Predicted continuous F0 / Hz



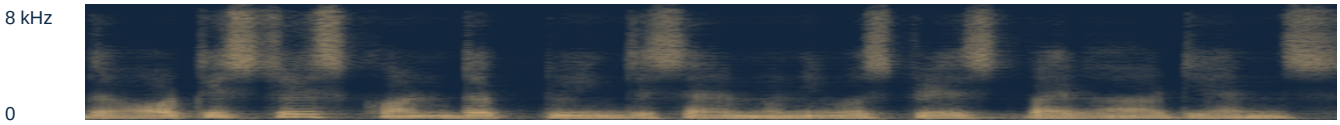
Predicted energy / STFT-magnitude units



x-axis: token index; original labeled plots accompany the submission



Measured waveform and spectrogram (Hann 1024 / hop 128; dB relative to utterance peak)



Waveform measurement: RMS -17.9 dBFS; peak -2.6 dBFS; 0 full-scale samples. The spectrogram describes generated acoustic structure and does not establish correct words.

Noun-context conduct should be examined together with E2b. Contextual stress cannot be established from a sentence-average F0. Compare the actual phone/stress subsequence and listen to the word; first-pronunciation dictionary lookup is a known limitation.

Listening observation: pending. A reviewer must assess pronunciation, completeness and prosody for this exact WAV. Predicted continuous F0 includes interpolated unvoiced regions; token means are unweighted.

Open E2a audio | Raw text and prediction arrays

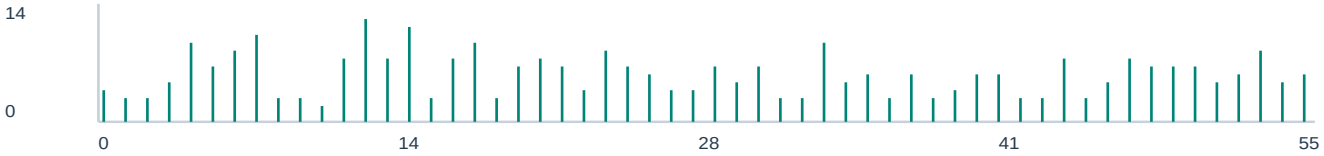
5 / E2b: per-sentence prosody analysis

English | step 8,000 | speaker ID 14

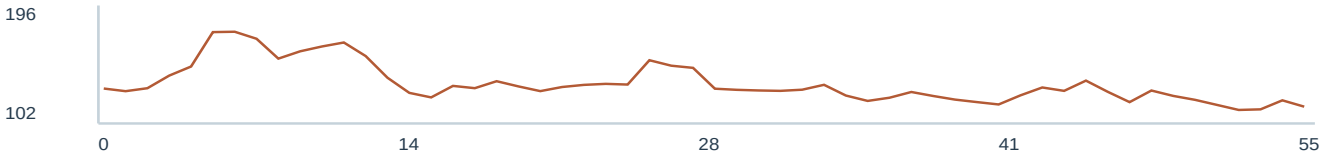
The researchers will conduct an experiment to evaluate the proposed method.

Saved measurement	Value
Tokens / expansion / duration	56 / 337 frames / 3.913 seconds
F0 token mean [range]	134.79 [112.90, 178.28] Hz
Energy token mean [range]	67.33 [9.44, 121.10]
Zero-duration tokens / explicit pause	0 tokens / 0 frames (0.000 s)

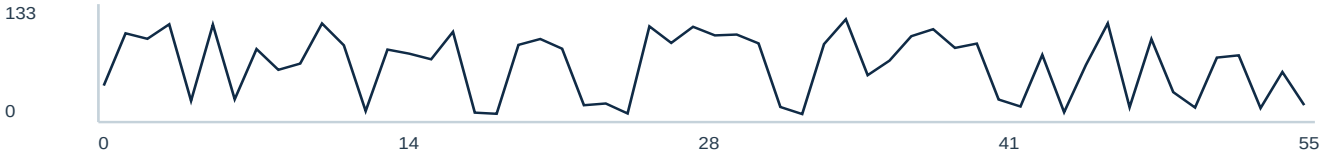
Duration / mel frames



Predicted continuous F0 / Hz



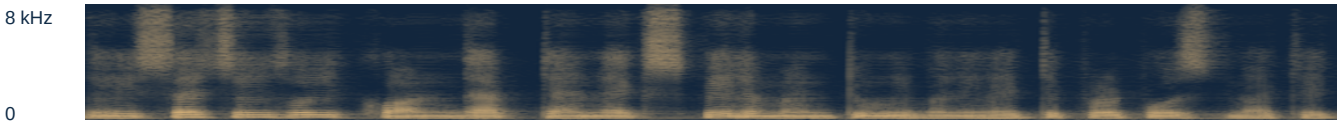
Predicted energy / STFT-magnitude units



x-axis: token index; original labeled plots accompany the submission



Measured waveform and spectrogram (Hann 1024 / hop 128; dB relative to utterance peak)



Waveform measurement: RMS -17.7 dBFS; peak -2.1 dBFS; 0 full-scale samples. The spectrogram describes generated acoustic structure and does not establish correct words.

The modal will supplies verb context for conduct. A contextual encoder can alter prosody but cannot reliably repair a wrong fixed pronunciation sequence. Pitch and duration differences from E2a must be interpreted alongside different surrounding words.

Listening observation: pending. A reviewer must assess pronunciation, completeness and prosody for this exact WAV. Predicted continuous F0 includes interpolated unvoiced regions; token means are unweighted.

Open E2b audio | Raw text and prediction arrays

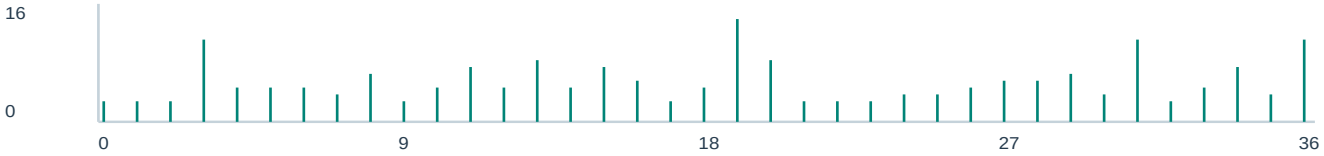
5 / E3: per-sentence prosody analysis

English | step 8,000 | speaker ID 14

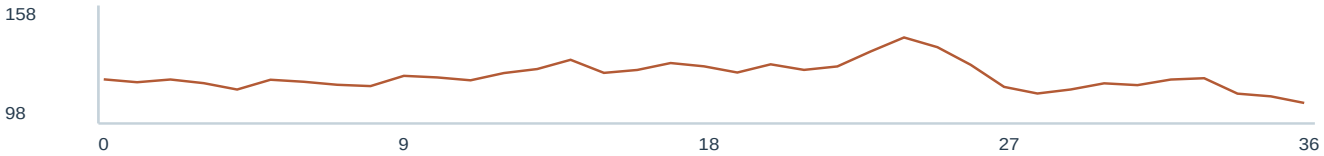
The rhyth m of the orchestra changed unexpectedly.

Saved measurement	Value
Tokens / expansion / duration	37 / 217 frames / 2.519 seconds
F0 token mean [range]	122.93 [108.71, 143.44] Hz
Energy token mean [range]	52.73 [6.44, 113.02]
Zero-duration tokens / explicit pause	0 tokens / 0 frames (0.000 s)

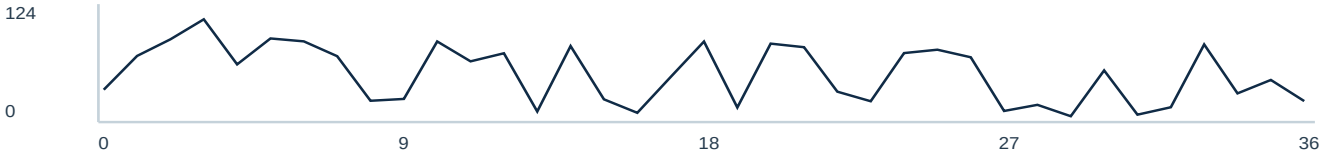
Duration / mel frames



Predicted continuous F0 / Hz



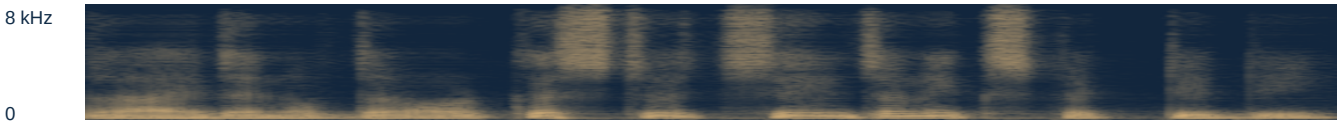
Predicted energy / STFT-magnitude units



x-axis: token index; original labeled plots accompany the submission



Measured waveform and spectrogram (Hann 1024 / hop 128; dB relative to utterance peak)



Waveform measurement: RMS -18.8 dBFS; peak -4.5 dBFS; 0 full-scale samples. The spectrogram describes generated acoustic structure and does not establish correct words.

The deliberately broken rhyth m input is retained. The frontend treats the pieces separately; the saved normalization collapses whitespace. Evaluate the pronunciation of both pieces without silently replacing the input with rhythm.

Listening observation: pending. A reviewer must assess pronunciation, completeness and prosody for this exact WAV. Predicted continuous F0 includes interpolated unvoiced regions; token means are unweighted.

Open E3 audio | Raw text and prediction arrays

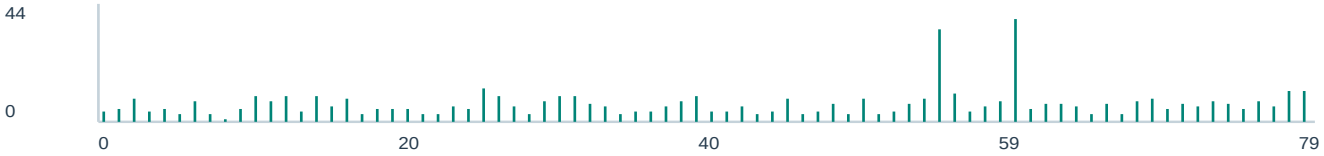
5 / E4: per-sentence prosody analysis

English | step 8,000 | speaker ID 14

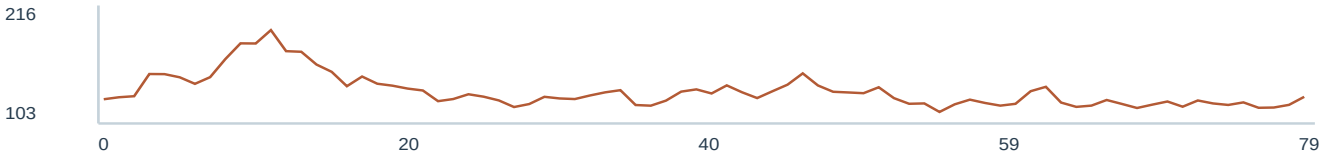
The meeting will start at 10:45 AM on 12 November 2026, so andaroo should arrive on time.

Saved measurement	Value
Tokens / expansion / duration	80 / 562 frames / 6.525 seconds
F0 token mean [range]	135.15 [114.78, 196.37] Hz
Energy token mean [range]	68.39 [5.23, 139.49]
Zero-duration tokens / explicit pause	0 tokens / 76 frames (0.882 s)

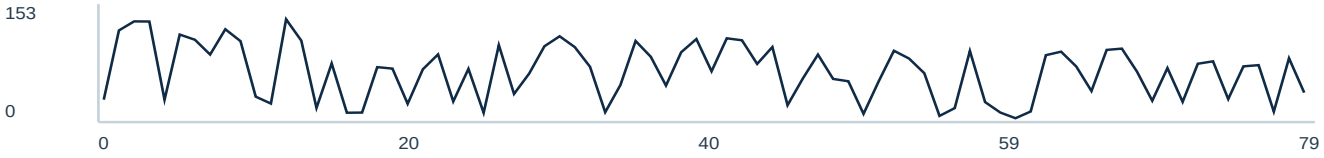
Duration / mel frames



Predicted continuous F0 / Hz



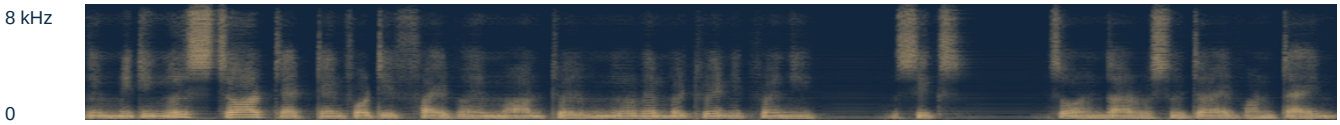
Predicted energy / STFT-magnitude units



x-axis: token index; original labeled plots accompany the submission



Measured waveform and spectrogram (Hann 1024 / hop 128; dB relative to utterance peak)



Waveform measurement: RMS -19.0 dBFS; peak -2.1 dBFS; 0 full-scale samples. The spectrogram describes generated acoustic structure and does not establish correct words.

Time/date expansion and andaroo require frontend review. The internal comma has an explicit sp token. Its duration is a model prediction, not a measurement of silence in the resulting waveform. Correct numeric words and loanword pronunciation need listening.

Listening observation: pending. A reviewer must assess pronunciation, completeness and prosody for this exact WAV. Predicted continuous F0 includes interpolated unvoiced regions; token means are unweighted.

Open E4 audio | Raw text and prediction arrays

5 / T1: per-sentence prosody analysis

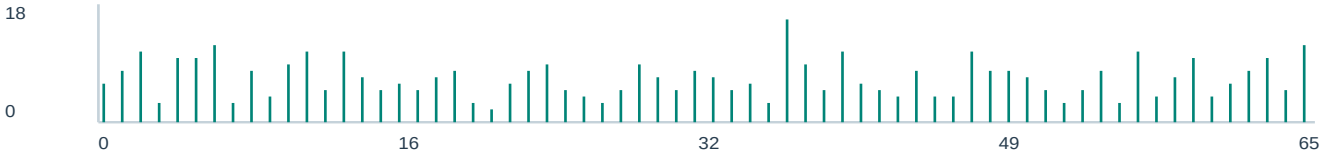
Telugu | step 13,000 | speaker ID 0

న్యూరోఫార్మకోలాజికల్ పరిశోధనలో నూతన ఔషధాల ప్రభావాన్ని విశ్లేషించారు.

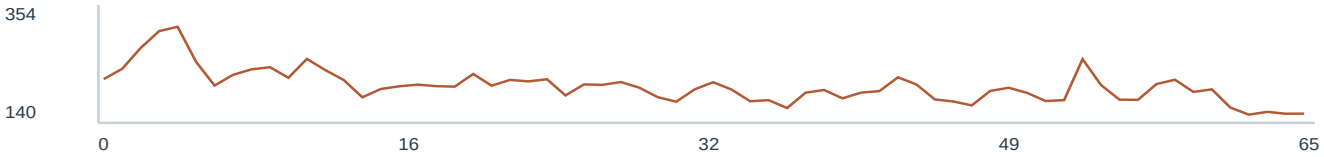
Exact UTF-8 source text is also preserved in the prediction JSON; the sentence above uses shaped Telugu rendering.

Saved measurement	Value
Tokens / expansion / duration	66 / 451 frames / 5.236 seconds
F0 token mean [range]	209.35 [155.40, 321.95] Hz
Energy token mean [range]	58.79 [8.16, 140.61]
Zero-duration tokens / explicit pause	0 tokens / 0 frames (0.000 s)

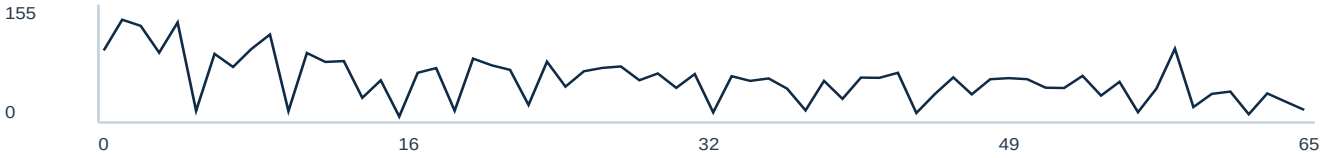
Duration / mel frames



Predicted continuous F0 / Hz



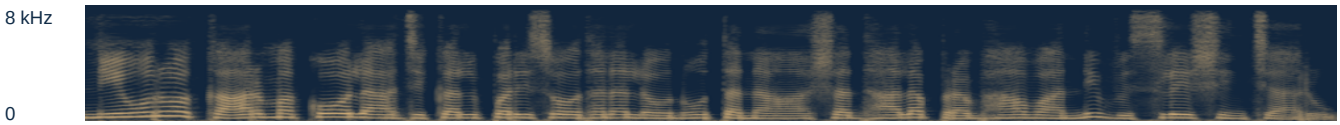
Predicted energy / STFT-magnitude units



x-axis: token index; original labeled plots accompany the submission



Measured waveform and spectrogram (Hann 1024 / hop 128; dB relative to utterance peak)



Waveform measurement: RMS -16.5 dBFS; peak -2.7 dBFS; 0 full-scale samples. The spectrogram describes generated acoustic structure and does not establish correct words.

The scientific loanword tests long consonant/vowel sequences and Telugu phonemic decomposition. Verify vowel length, clusters and complete delivery by a Telugu listener. A smooth predicted F0 curve is not independent evidence of accurate pronunciation.

Listening observation: pending. A reviewer must assess pronunciation, completeness and prosody for this exact WAV. Predicted continuous F0 includes interpolated unvoiced regions; token means are unweighted.

Open T1 audio | Raw text and prediction arrays

5 / T2a: per-sentence prosody analysis

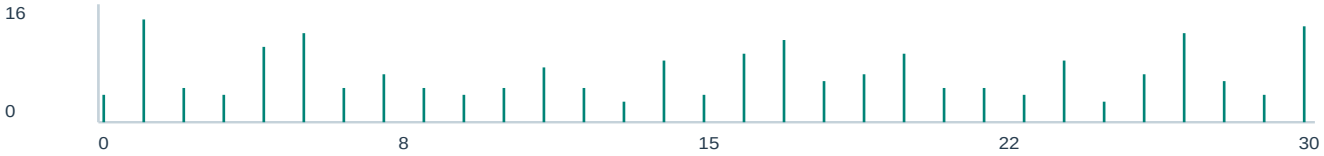
Telugu | step 13,000 | speaker ID 0

తాడుపై అతని పట్టు చాలా బలంగా ఉంది.

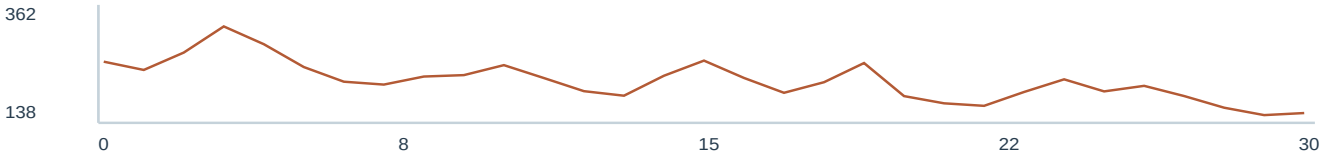
Exact UTF-8 source text is also preserved in the prediction JSON; the sentence above uses shaped Telugu rendering.

Saved measurement	Value
Tokens / expansion / duration	31 / 222 frames / 2.577 seconds
F0 token mean [range]	221.29 [153.51, 329.22] Hz
Energy token mean [range]	59.73 [4.07, 130.15]
Zero-duration tokens / explicit pause	0 tokens / 0 frames (0.000 s)

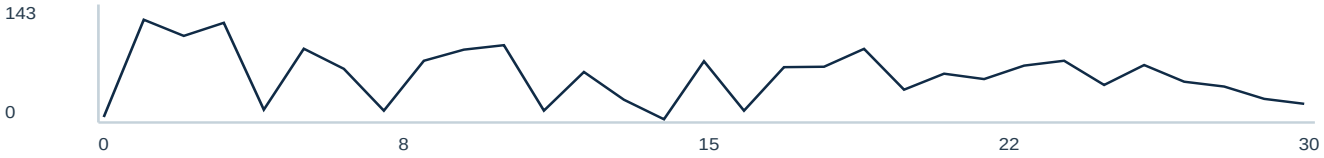
Duration / mel frames



Predicted continuous F0 / Hz



Predicted energy / STFT-magnitude units



x-axis: token index; original labeled plots accompany the submission



Measured waveform and spectrogram (Hann 1024 / hop 128; dB relative to utterance peak)



Waveform measurement: RMS -16.6 dBFS; peak -3.2 dBFS; 0 full-scale samples. The spectrogram describes generated acoustic structure and does not establish correct words.

Pattu occurs in the grip/hold context. Inspect the two retroflex consonant phones and their durations for the intended gemination. Phone durations alone do not establish that a natural closure and release were realized by the vocoder.

Listening observation: pending. A reviewer must assess pronunciation, completeness and prosody for this exact WAV. Predicted continuous F0 includes interpolated unvoiced regions; token means are unweighted.

Open T2a audio | Raw text and prediction arrays

5 / T2b: per-sentence prosody analysis

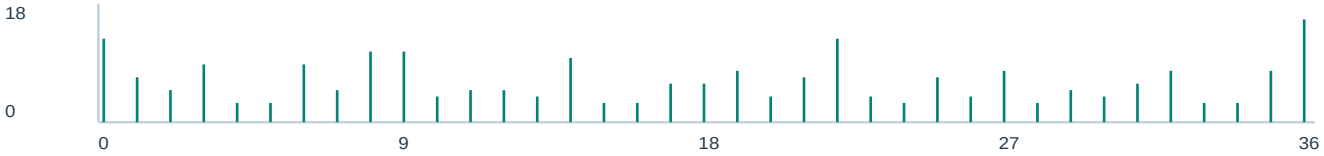
Telugu | step 13,000 | speaker ID 0

ఆమె పట్టు చీరను ధరించి వేడుకకు వచ్చింది.

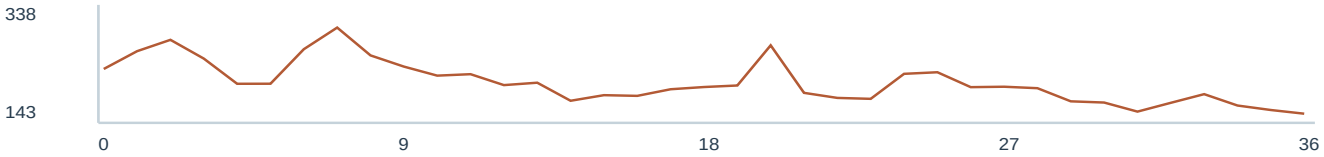
Exact UTF-8 source text is also preserved in the prediction JSON; the sentence above uses shaped Telugu rendering.

Saved measurement	Value
Tokens / expansion / duration	37 / 236 frames / 2.740 seconds
F0 token mean [range]	213.02 [158.50, 307.49] Hz
Energy token mean [range]	55.04 [5.65, 133.13]
Zero-duration tokens / explicit pause	0 tokens / 0 frames (0.000 s)

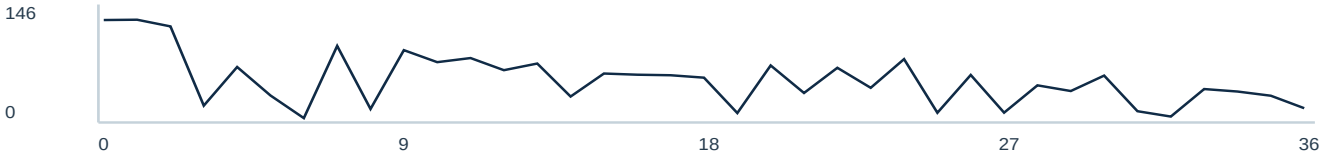
Duration / mel frames



Predicted continuous F0 / Hz



Predicted energy / STFT-magnitude units



x-axis: token index; original labeled plots accompany the submission



Measured waveform and spectrogram (Hann 1024 / hop 128; dB relative to utterance peak)



Waveform measurement: RMS -17.1 dBFS; peak -3.7 dBFS; 0 full-scale samples. The spectrogram describes generated acoustic structure and does not establish correct words.

Pattu occurs before the noun for sari in the silk context. The same spelling can have similar segmental pronunciation with different context-dependent prosody. There is no requirement that T2a and T2b have identical F0 or duration.

Listening observation: pending. A reviewer must assess pronunciation, completeness and prosody for this exact WAV. Predicted continuous F0 includes interpolated unvoiced regions; token means are unweighted.

Open T2b audio | Raw text and prediction arrays

5 / T3: per-sentence prosody analysis

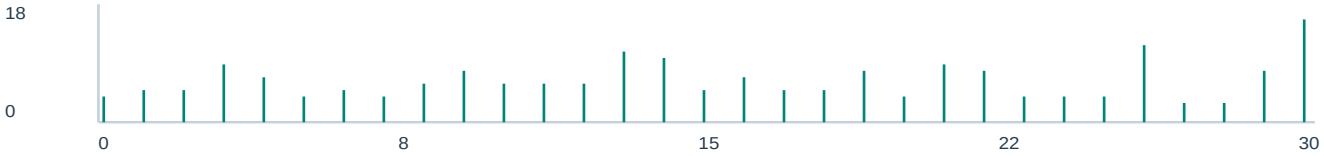
Telugu | step 13,000 | speaker ID 0

పులి అడ విలో నిశ్శబ్దంగా నడిచింది.

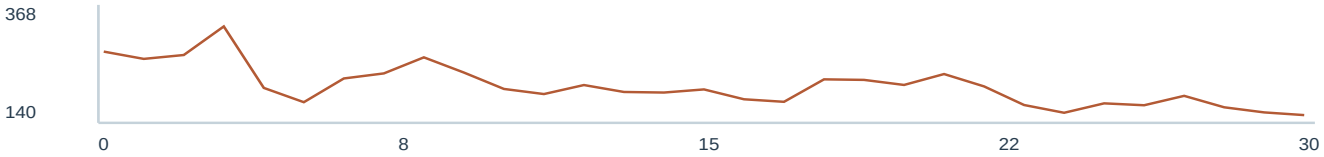
Exact UTF-8 source text is also preserved in the prediction JSON; the sentence above uses shaped Telugu rendering.

Saved measurement	Value
Tokens / expansion / duration	31 / 201 frames / 2.334 seconds
F0 token mean [range]	214.33 [155.94, 334.33] Hz
Energy token mean [range]	63.36 [5.64, 115.50]
Zero-duration tokens / explicit pause	0 tokens / 0 frames (0.000 s)

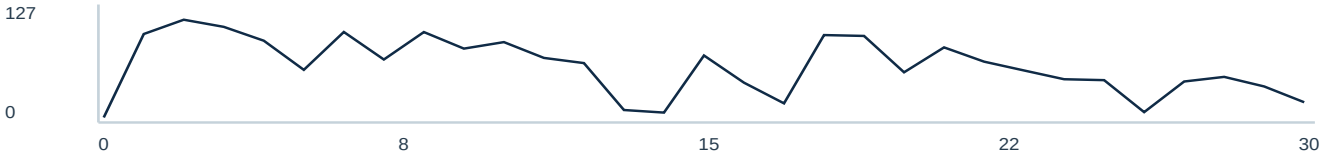
Duration / mel frames



Predicted continuous F0 / Hz



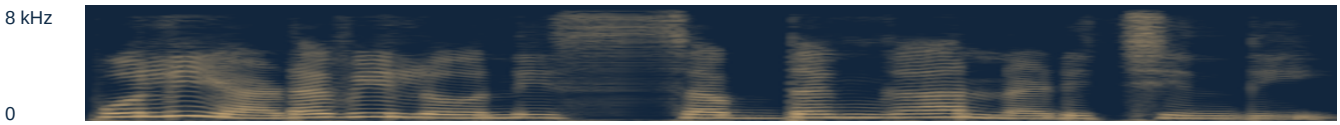
Predicted energy / STFT-magnitude units



x-axis: token index; original labeled plots accompany the submission



Measured waveform and spectrogram (Hann 1024 / hop 128; dB relative to utterance peak)



Waveform measurement: RMS -16.5 dBFS; peak -3.1 dBFS; 0 full-scale samples. The spectrogram describes generated acoustic structure and does not establish correct words.

The broken ada vilo word and double space are intentional raw inputs. Analyze unwanted pauses, changed syllabification and geminate realization by listening; do not automatically join the fragments when evaluating the required output.

Listening observation: pending. A reviewer must assess pronunciation, completeness and prosody for this exact WAV. Predicted continuous F0 includes interpolated unvoiced regions; token means are unweighted.

Open T3 audio | Raw text and prediction arrays

5 / T4: per-sentence prosody analysis

Telugu | step 13,000 | speaker ID 0

ఈ సమావేశం 2026 నవంబర్ 12వ తేదీ ఉదయం 10:45 గంటలకు ప్రారంభమవుతుంది, సమయానికి సిద్ధంగా ఉంచండి.

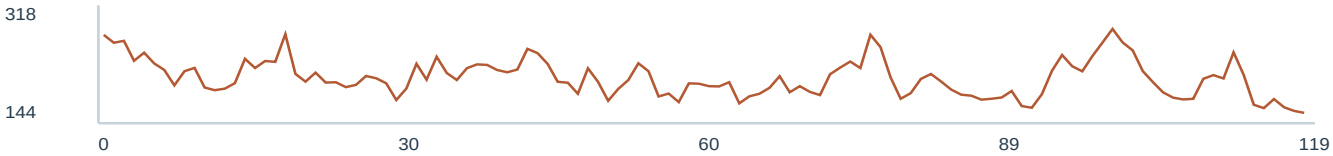
Exact UTF-8 source text is also preserved in the prediction JSON; the sentence above uses shaped Telugu rendering.

Saved measurement	Value
Tokens / expansion / duration	120 / 775 frames / 8.998 seconds
F0 token mean [range]	213.89 [160.12, 289.32] Hz
Energy token mean [range]	56.93 [1.38, 114.45]
Zero-duration tokens / explicit pause	0 tokens / 52 frames (0.604 s)

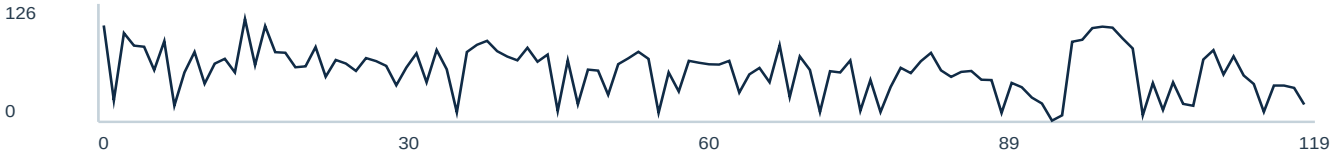
Duration / mel frames



Predicted continuous F0 / Hz



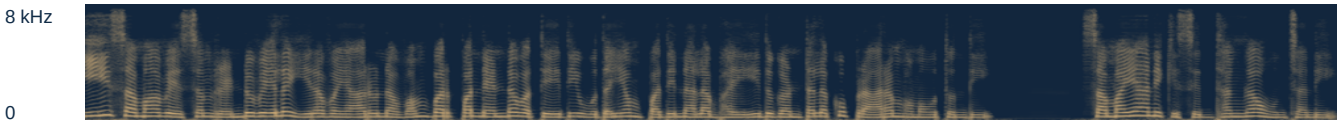
Predicted energy / STFT-magnitude units



x-axis: token index; original labeled plots accompany the submission



Measured waveform and spectrogram (Hann 1024 / hop 128; dB relative to utterance peak)



Waveform measurement: RMS -17.1 dBFS; peak -3.3 dBFS; 0 full-scale samples. The spectrogram describes generated acoustic structure and does not establish correct words.

This is the longest Telugu test output and exercises year, ordinal and time expansion. Inspect normalization of 2026, 12va and 10:45, sentence completeness and the comma pause. Numeric expansion can produce grammatical or pronunciation errors even when all token IDs are valid.

Listening observation: pending. A reviewer must assess pronunciation, completeness and prosody for this exact WAV. Predicted continuous F0 includes interpolated unvoiced regions; token means are unweighted.

Open T4 audio | Raw text and prediction arrays

6 / Comparison, limitations and quality review

Dimension	English	Telugu
Source character	Accent-diverse ASR corpus; multiple demographic profiles	Studio TTS corpus; two source voices
Submitted synthesis	Step 8,000; speaker 14	Step 13,000; speaker 0
Frontend risk	OOVs, conduct stress, dates and loanwords	Clusters, vowel length, anusvara and number grammar
Interpretation	Cannot separate accent/noise/speaker effects from model performance	Lower visual losses do not independently prove superior speech

Listening protocol required for final quality claims

Listen to all ten WAVs at normal speed with headphones; use a Telugu-proficient reviewer for T1-T4. Record intelligibility, omitted/repeated content, target-word pronunciation, prosody and clipping. Compare real-mel vocoder reconstruction first to separate vocoder mismatch from acoustic-model failure. Compare reference-conditioned and free-running held-out samples to isolate variance-prediction errors. Keep the exact input text, speaker and selected checkpoint fixed.

No unmeasured results

This report does not claim MOS, ASR WER, speaker similarity, stress correctness or natural pronunciation without measurements/listening records. The saved artifacts establish waveform existence and predictor values. Speech quality is still an open evaluation item until reviewed. The submission checker verifies structure, text and numerical consistency, not perceptual quality.

Prioritized improvements

Selected local checkpoints and source logs are available. First recover missing reserve hashes and inspect the 82 Telugu target mismatches. Review every WAV using the provided audio reviewer, then diagnose alignments/speakers and choose new checkpoints on held-out free-running speech. Only after these checks consider stronger alignment models, context-sensitive pronunciations, richer pitch modeling or vocoder adaptation. More training is useful only when held-out behavior improves.

Assignment completion status

The structural audit checks ten WAVs, thirty variance plots, four loss images, raw-text equality, predictor lengths, duration sums and audio format. It currently reports 4 missing-evidence items. Refer to Results/submission_audit.json and model_artifacts.json; do not call the project fully reproducible while these items remain unresolved.

- English: audio hashes missing for 382 retained reserve utterances; complete final audio-overlap verification is pending
- English: per-sentence listening observations pending: E1, E2a, E2b, E3, E4
- Telugu: audio hashes missing for 339 retained reserve utterances; complete final audio-overlap verification is pending
- Telugu: per-sentence listening observations pending: T1, T2a, T2b, T3, T4

7 / Reproducibility and evaluator access

Selected model bundles

Owner-supplied artifact folder: Google Drive model and experiment artifacts. Historical artifact links are recorded in model_artifacts.json. Local selected checkpoint bytes match the registry. Public Drive access was not rechecked in this network-restricted revision.

English / step 8,000: download checkpoint

Telugu / step 13,000: download checkpoint

Restore the published small metadata bundle with python tools/restore_artifacts.py. Add --download for missing checksum-verified acoustic/vocoder weights; Drive may require a manual browser download. Metadata is under Results/Language/metadata; source snapshots are labeled as recovered evidence, not a proven training commit.

Inference after restoring the matching model bundle

```
python English/inference.py --config English/config.yaml
  --ckpt_file English/checkpoints/8000.pth.tar
  --speaker_id 14 --result_path experiments/review/English
```

```
python Telugu/inference.py --config Telugu/config.yaml
  --ckpt_file Telugu/checkpoints/13000.pth.tar
  --speaker_id 0 --result_path experiments/review/Telugu
```

Run from the repository root; join wrapped command lines. Configuration paths resolve relative to their YAML file. Use new result directories to preserve supplied audio. Fresh alignment, feature validation, 64-clip diagnostic, held-out pilots and optional full training are implemented in the Colab notebook.

Environment and evidence

The acoustic requirements file uses lower version bounds; it is not an environment lock. Export the actual training environment and CUDA/PyTorch versions with the original logs. Report generation uses reportlab, Pillow and pypdfium2. Rebuild with python Report/generate_report.py; audit with python tools/audit_submission.py --strict.

References

[1] Assignment: FastSpeech2_Assignment.pdf, sections 1-4, including all seven technical-report requirements.

[2] Ren et al. FastSpeech 2: Fast and High-Quality End-to-End Text to Speech. Supplied arXiv:2006.04558v8. https://arxiv.org/abs/2006.04558

[3] ming024/FastSpeech2. Reference commit d4e79eb52e8b01d24703b2dfc0385544092958f3. https://github.com/ming024/FastSpeech2

[4] Dataset sources: https://huggingface.co/datasets/ai4bharat/Svarah and https://huggingface.co/datasets/SPRINGLab/IndicTTS_Telugu

[5] Repository evidence: language configurations, preprocessing/model/frontend source, Results/*/variance_predictions_summary.json and four loss PNGs.